SPEAKER 1 OF 4 — PROBLEM FIT & TECHNICAL FEASIBILITY · 40 MARKS

Updesh Singh

Opening, the problem, and our technology

SAY THIS

Hello sir. Myself **Updesh Singh**, and we are **Team Deviar 3**. We are presenting our project **Samarth Sentinel**, for problem statement DEV-104 from Samarth Life Management.

Sir, Samarth supports **40,000 seniors across 350 cities**. Each Care Coordinator looks after many families and can realistically call about **ten people a day**. Today they pick those ten from memory and old call notes.

So we asked one question. Is the real problem *predicting illness*? We decided it is not. The real problem is — **"which ten people do I call this morning, and what do I say when they pick up?"**

That is a ranking problem under a hard daily limit, and this one decision shaped everything after it. We measure **how good the top of the list is**, not overall accuracy. A model that ranks well but sends fifty alerts is useless to the person holding the phone.

Sir, on the technical side — this is not a slide deck. It is a **working product, deployed live**. You can open it on your phone right now.

The frontend is **Next.js 15 with TypeScript**. Data and login are on **Supabase PostgreSQL**. The model is **Python with scikit-learn** — gradient boosting plus an isolation forest. Speech-to-text for Hindi uses **Whisper**, and the weekly family note is written by an **LLM**.

And sir, the whole thing runs on **free cloud tiers with no GPU**. That matters, because the brief asks it to stay affordable at 10,000 seniors.

TECH STACK

Next.js 15 · React 19 | **TypeScript** | **Supabase** · PostgreSQL | Row-Level Security | Auth + Google sign-in | **Python 3.11** | **scikit-learn** | Isolation Forest | **Whisper** · speech-to-text | LLM · family summaries | Web Speech · text-to-speech | **Vercel** · deployed live

150M+
Indians aged 60+, many living alone

~10 a day
what one coordinator can act on

Free tier
no GPU, runs on modest cloud

BHAI, SIMPLE BAAT

Tereko **do cheez** batani hain — problem kya hai, aur banaya kis cheez se hai.

Samarth ke paas 40,000 buzurg hain, par ek coordinator din mein sirf 10 ko call kar sakta hai. Abhi wo yaadasht se decide karta hai. Yahi dikkat hai.

Toh asli sawaal ye nahi ki "kaun bimaar padega". Asli sawaal hai — **"aaj subah main kin 10 logon ko phone karun?"** Ye line poore project ka base hai, confidently bolna.

Phir tech bolna — Next.js frontend, Supabase database, Python ka model, Whisper se Hindi awaaz samajhna. Aur sabse strong line: **"sir ye live chal raha hai, abhi phone pe khol ke dikha sakte hain"**. Aur ye bhi — poora system **free cloud pe, bina GPU** ke chalta hai, matlab sasta hai.

Next — **"Sir, ab meri teammate Anamika batayengi ki system kaam kaise karta hai."**

SPEAKER 2 OF 4 — INNOVATION & ORIGINALITY · 20 MARKS

Anamika

How the system works, and what is new in it

SAY THIS

Good morning sir. I am **Anamika**. Let me explain how the system actually works.

Every senior gets a **daily risk score**, built from four layers — simple clinical rules, the senior's *own* personal baseline, a machine learning ranker, and an anomaly detector.

Sir, the second layer is the important one. We do not compare a senior against the general population. A blood pressure of 150 can be completely normal for one person and an emergency change for another. So we compare **each senior only against their own past readings**.

Three things make our approach different. First, **we show what we held back**. The list is capped at ten, and the seniors we scored but did not alert on appear greyed out below the line. A coordinator who cannot see what was withheld has no reason to trust the list.

Second, **silence is evidence**. If a senior does not answer the daily call, we do not leave it blank. An unanswered call is itself a warning, and it is scored.

Third, **every alert is a sentence, not a number**. Instead of "risk 0.64", the coordinator reads "under 57% of doses logged this week", and "call this morning and go through the evening routine".

Sir, one more thing. Indian call notes are mixed Hindi and English — *"thoda dard hai"*. We built an extractor that reads that and pulls out **eight types of health cue**. And it understands negation: *"gire nahi"* means they did *not* fall, so we raise nothing.

4 layers
rules + personal baseline + ML + anomaly

8 cues
pulled from mixed Hindi-English notes

Capped at 10
and we show what was held back

BHAI, SIMPLE BAAT

Tereko batana hai ki **system kaam kaise karta hai** aur **isme naya kya hai**.

Har buzurg ka roz ka ek score banta hai, 4 alag cheezon ko milakar. Sabse badi baat — hum kisi ko doosre se compare nahi karte, **usi insaan ke apne purane readings se** compare karte hain. BP 150 kisi ke liye normal hai, kisi ke liye khatra. Yahi cheez judge ko sabse zyada impress karegi.

Teen cheezein humari alag hain: **(1)** jin logon ko humne jaanbujhkar alert nahi kiya, unke naam bhi dikhate hain — taaki coordinator ko bharosa rahe ki kuch chhupaya nahi. **(2)** agar buzurg phone nahi uthata toh wo bhi signal hai, khali nahi chhodte. **(3)** alert mein sirf number nahi, poori baat likhi hoti hai ki karna kya hai.

Aur last mein ye bolna — call notes Hindi-English mix mein hote hain, jaise "thoda dard hai". Humne wo padhne wala system banaya. Aur agar likha ho "gire nahi" toh matlab giray nahi — system ye samajh jaata hai, warna galat alert chala jaata.

Next — **"Sir, ab Riddhima batayengi ki data kaise banaya aur privacy kaise sambhaali."**

SPEAKER 3 OF 4 — DATA, RESEARCH QUALITY & PRIVACY · 30 MARKS

Riddhima

How we built our data, and how we protect it

SAY THIS

Good morning sir. I am **Riddhima**. I will cover two things — how we built our data, and how we protect it.

Sir, no real patient data exists for this. So we generated a synthetic cohort of **200 seniors over 90 days** — that is **18,000 senior-days**.

But sir, we did not invent the numbers randomly. We took **real anonymised health records from NHANES**, a public government health survey, and drew **whole people** from it — so blood pressure, weight and sugar stay realistically related to each other. If you generate each number separately, you create people who could not exist.

Inside that data we hid **24 decline cases** in three different patterns — gradual, sudden, and on-and-off. And this is the part I am most proud of: we also planted **49 cases that look like decline but are not** — a flu week, a travel week with missed calls, and a faulty BP machine. Without those, a system that simply alerts on everything would score well and be useless.

Sir, now **privacy**. A family member cannot see blood pressure, sugar, or call notes. But not because we wrote an instruction in our code to hide them.

There is **no database permission granting them access at all**. So even if our website had a bug, the database returns zero rows. We tested this signing in as each role — family, operations, and a logged-out visitor. All blocked.

And when we use an AI service to write the family note, **we never send the senior's real name** — we send a placeholder and put the name back on our own machine.

200 × 90
seniors × days — 18,000 senior-days

24 hidden
decline cases, three different patterns

49 traps
look like decline, deliberately are not

BHAI, SIMPLE BAAT

Tereko **do cheez** batani hain — data kaise banaya, aur privacy kaise handle ki.

Asli mareezon ka data milta nahi hai, toh humne khud banaya — 200 buzurg, 90 din ka. Par randomly nahi banaya. **NHANES** naam ka ek sarkari health survey hai, usme se poore-poore insaan uthaye, taaki BP, weight, sugar aapas mein sahi tareeke se jude rahein.

Phir usme 24 case chhupaye jahan tabiyat bigad rahi thi. Aur sabse smart cheez — **49 aise case bhi daale jo bigadte hue lagte hain par actually normal hain** (jaise flu ka hafta, ya BP machine kharab). Agar ye na daalte toh koi bhi faltu system bhi acha score kar leta. Ye point zarur bolna, judge ko accha lagega.

Privacy wali baat — family ko BP, sugar, call notes kuch nahi dikhta. Par log sochte hain humne code mein chhupaya hoga. Nahi. **Database mein unko permission hi nahi di.** Matlab website mein bug bhi aa jaye toh bhi kuch nahi dikhega. Isko zor dekar bolna.

Next — **"Sir, ab Bhanu hamare results aur impact batayenge."**

SPEAKER 4 OF 4 — RESULTS, USER NEEDS & IMPACT · 10 MARKS

Bhanu

Results, who it helps, and closing

SAY THIS

Good morning sir. I am **Bhanu**, and I will close with our results.

Sir, those **24 hidden decline cases** Riddhima mentioned — our system caught **all 24**, and caught every one of them **before the day it became obvious**. Median **22 days early**.

On precision I want to be honest. Our number is **0.169**. But sir, please look at the ceiling first. With 24 real cases spread over 90 days and only ten slots a day, the *highest mathematically possible* precision is **0.192**. Most days genuinely have fewer than ten declining seniors. So we reach **88% of what is even achievable**.

The coordinator gets a short ordered list with a reason and an action — something to act on tomorrow morning, not a dashboard to study.

The senior needs no app and no smartphone. Six short questions on a normal phone call, answered by pressing 1, 2 or 3, in Hindi or English — on the handset they already own.

The family gets a warm weekly note in plain language, no readings and no scores. They can write back to the coordinator too — because a daughter knows things no sensor records.

Sir, one thing did **not** work. We built the feedback loop where coordinators mark alerts useful or not. We measured it, and it improved precision by almost nothing. We are reporting that honestly rather than hiding it.

Samarth Sentinel never diagnoses. It notices a change, explains it in plain words, and puts it in front of a human who can pick up the phone. **Thank you, sir.**

24 / 24
hidden declines caught, all before the obvious day

22 days
median early warning

88%
of the achievable precision

BHAI, SIMPLE BAAT

Tereko **result** batane hain aur **kis-kis ko fayda hai** wo.

Jo 24 case Riddhima ne chhupaye the, system ne **saare 24 pakad liye** — aur har ek ko pehle hi, average 22 din pehle. Ye sabse strong point hai, isko sabse pehle bolna.

Precision 0.169 hai — sunne mein kam lagta hai, judge pooch sakta hai. Toh **pehle ceiling bolna, number baad mein**: "sir maximum possible hi 0.192 hai, hum 88% tak pahunche". Ulta bologe toh kam lagega.

Phir teen log — coordinator ko chhoti list with reason, buzurg ko sirf phone call (app ki zarurat nahi, 1-2-3 dabana hai bas), aur family ko simple bhasha mein note bina kisi medical reading ke.

Aur last mein ek cheez jo kaam nahi kari wo bhi bolna — feedback loop se improvement nahi aaya. Lagta hai galat bolna, par **isse impression accha padta hai**, kyunki iska matlab humne sach mein measure kiya, guess nahi kiya.